

HUDCO Trainee Officer

Previous Year Paper
(Electrical / Electronics)
20 Oct, 2024 Shift 3

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Participant ID	
Participant Name	
Test Center Name	
Test Date	20/10/2024
Test Time	4:30 PM - 6:30 PM
Subject	Trainee Officer Projects(Electrical or Electrical and Electronics)

Section : General English, Quantitative Aptitude, Reasoning and General Awareness

Q.1 With respect to the Union Council of Ministers, before a Minister enters upon his office, the President shall administer to him the oaths of office and of secrecy according to the forms set out for the purpose in the _____ Schedule of the Indian Constitution.

- Ans
- ☒ 1. Seventh
 - ☒ 2. Third
 - ☒ 3. Fourth
 - ☒ 4. Fifth

Question ID : 6306801000884
Option 1 ID : 6306803927585
Option 2 ID : 6306803927582
Option 3 ID : 6306803927583
Option 4 ID : 6306803927584
Status : Answered
Chosen Option : 3

Q.2 Select the set in which the numbers are related in the same way as are the numbers of the following sets.
(NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g., 13 – Operations on 13 such as adding/deleting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed)
(14,198,693)
(13,12, 39)

- Ans
- ☒ 1. (7, 51, 89)
 - ☒ 2. (9, 83, 182)
 - ☒ 3. (8, 66, 132)
 - ☒ 4. (5, 27, 38)

Question ID : 630680517092
Option 1 ID : 6306802020767
Option 2 ID : 6306802020765
Option 3 ID : 6306802020766
Option 4 ID : 6306802020764
Status : Answered
Chosen Option : 2

Q.3 If the top of the two towers of height h_1 and h_2 units, standing on level ground subtend angles of 45° and 60° , respectively, at the center of the line joining their feet, then $h_1^2 : h_2^2$ is:

- Ans
- ☒ 1. $1 : \sqrt{3}$
 - ☒ 2. $1 : 3$
 - ☒ 3. $\sqrt{3} : 1$
 - ☒ 4. $3 : 1$

Question ID : 630680374132
Option 1 ID : 6306801457800
Option 2 ID : 6306801457801
Option 3 ID : 6306801457799
Option 4 ID : 6306801457802
Status : Not Answered
Chosen Option : --

Q.4 Three of the following four letter-clusters are alike in a certain way and thus form a group. Which is the letter-cluster that does NOT belong to that group?
(Note: The odd one out is not based on the number of consonants/vowels or their positions in the letter-cluster.)

- Ans
- ☒ 1. QTZ
 - ☒ 2. HKN
 - ☒ 3. ORX
 - ☒ 4. CFL

Question ID : 630680723556
Option 1 ID : 6306802835421
Option 2 ID : 6306802835420
Option 3 ID : 6306802835419
Option 4 ID : 6306802835418
Status : Answered
Chosen Option : 2

Q.5 As per the Indian Constitution, who among the following is the Chairman of the Inter-State Council of India?

- Ans
- ☒ 1. President of India
 - ☒ 2. Cabinet Secretary
 - ☒ 3. Prime Minister of India
 - ☒ 4. Ministry of Home Affairs

Question ID : 6306801000895
Option 1 ID : 6306803927626
Option 2 ID : 6306803927628
Option 3 ID : 6306803927627
Option 4 ID : 6306803927629
Status : Answered
Chosen Option : 2

Q.6 Parts of the following sentence have been given as options. Select the option that contains an error.
Thirty years have passed that she came back from Manhattan.

- Ans
- ☒ 1. Thirty years
 - ☒ 2. have passed
 - ☒ 3. from Manhattan
 - ☒ 4. that she came back

Question ID : 630680154852
Option 1 ID : 630680599204
Option 2 ID : 630680599205
Option 3 ID : 630680599207
Option 4 ID : 630680599206
Status : Answered
Chosen Option : 4

Q.7 How many meaningful four-letter English words can be formed using J, E, A and R using all letters but each letter only once in each word?

- Ans
- ☒ 1. Two
 - ☒ 2. One
 - ☒ 3. Zero
 - ☒ 4. More than two

Question ID : 630680151193
Option 1 ID : 630680584834
Option 2 ID : 630680584833
Option 3 ID : 630680584832
Option 4 ID : 630680584835
Status : Answered
Chosen Option : 3

Q.8 Which of the following trees is characteristic of the alpine zone of the Western Himalayan region?

- Ans
- ☒ 1. Rhododendron
 - ☒ 2. Oak
 - ☒ 3. Maples
 - ☒ 4. Juniper

Question ID : 630680741916
Option 1 ID : 6306802908255
Option 2 ID : 6306802908252
Option 3 ID : 6306802908254
Option 4 ID : 6306802908253
Status : Answered
Chosen Option : 1

Q.9 Sentences of a paragraph are given below in jumbled order. Arrange the sentences in the correct order to form a meaningful and coherent paragraph.

- (i) However, the COVID-19 pandemic led India's economy into a contraction of 7.3 per cent in FY21, despite well-crafted fiscal and monetary policy support.
(ii) Following the deadly 'second wave', growth in FY22 is expected to be nearer to the lower bound of the range of 7.5 to 12.5 per cent – still putting India among the fastest growing economies in the world.
(iii) Since the 2000s, India has made remarkable progress in reducing absolute poverty.
(iv) Between 2011 and 2015, more than 90 million people were lifted out of extreme poverty.

- Ans ☒ 1. (iii), (i), (iv), (ii)
☒ 2. (iii), (i), (ii), (iv)
☒ 3. (iii), (iv), (i), (ii)
☒ 4. (i), (ii), (iii), (iv)

Question ID : 630680102573
Option 1 ID : 630680398554
Option 2 ID : 630680398553
Option 3 ID : 630680398552
Option 4 ID : 630680398551
Status : Answered
Chosen Option : 2

Q.10 Which Article of the Indian Constitution provides for the protection in respect of conviction for offences?

- Ans ☒ 1. Article 19
☒ 2. Article 21
☒ 3. Article 22
☒ 4. Article 20

Question ID : 6306801031716
Option 1 ID : 6306804049561
Option 2 ID : 6306804049563
Option 3 ID : 6306804049564
Option 4 ID : 6306804049562
Status : Answered
Chosen Option : 1

Q.11 The LCM of two numbers is 2112 and their HCF is 66. If one of the two numbers is 121, find the other number.

- Ans ☒ 1. 1152
☒ 2. 1113
☒ 3. 1162
☒ 4. 1142

Question ID : 630680796091
Option 1 ID : 6306803119352
Option 2 ID : 6306803119355
Option 3 ID : 6306803119354
Option 4 ID : 6306803119353
Status : Answered
Chosen Option : 1

Q.12 The first flight under 'PM Shri Paryatan Vayu Seva', which was launched by Madhya Pradesh Chief Minister Mohan Yadav in June 2024 to promote tourism in the state, flagged off from _____.

- Ans ☒ 1. Bhopal to Jabalpur
☒ 2. Bhopal to Gwalior
☒ 3. Bhopal to Indore
☒ 4. Bhopal to Ujjain

Question ID : 6306801018363
Option 1 ID : 6306803996980
Option 2 ID : 6306803996981
Option 3 ID : 6306803996979
Option 4 ID : 6306803996982
Status : Answered
Chosen Option : 4



Q.13 Who among the following founded the Ramakrishna Mission and was a key proponent of Vedanta and Hindu unity?

- Ans
- ✗

1. Rabindranath Tagore

✗

2. Satyendranath Tagore

✗

3. Raja Ram Mohan Roy

✓

4. Swami Vivekananda

Question ID : 6306801057020

Option 1 ID : 6306804150507

Option 2 ID : 6306804150508

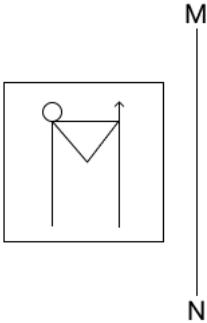
Option 3 ID : 6306804150506

Option 4 ID : 6306804150505

Status : Answered

Chosen Option : 4

Q.14 Select the correct mirror image of the given figure when the mirror is placed at MN as shown below.



- Ans
- ✗

1.

✗

2.

✓

3.

✗

4.

Question ID : 630680624542

Option 1 ID : 6306802444342

Option 2 ID : 6306802444344

Option 3 ID : 6306802444341

Option 4 ID : 6306802444343

Status : Answered

Chosen Option : 3

Q.15 What should come in place of ? in the given series in order to make it logically complete?

PQ20 MN15 JK10 ?

- Ans
- ☐ 1. HG10
 - ☐ 2. GH10
 - ☐ 3. HG05
 - ☒ 4. GH05

Question ID : 630680366885
Option 1 ID : 6306801429010
Option 2 ID : 6306801429012
Option 3 ID : 6306801429011
Option 4 ID : 6306801429009
Status : Answered
Chosen Option : 4

Q.16 Select the most appropriate ANTONYM of the given word.

Ameliorist

- Ans
- ☐ 1. Reformist
 - ☐ 2. Pugilist
 - ☐ 3. Populist
 - ☒ 4. Reactionary

Question ID : 630680996674
Option 1 ID : 6306803910431
Option 2 ID : 6306803910428
Option 3 ID : 6306803910429
Option 4 ID : 6306803910430
Status : Answered
Chosen Option : 2

Q.17 Simplify:

$$(3z - 7y)^2 + (7z + 3y)^2 - 49z^2$$

- Ans
- ☒ 1. $9z^2 + 58y^2$
 - ☐ 2. $13z^2 + 50y^2$
 - ☐ 3. $14z^2 + 60y^2$
 - ☐ 4. $6z^2 + 50y^2$

Question ID : 630680704744
Option 1 ID : 6306802760662
Option 2 ID : 6306802760664
Option 3 ID : 6306802760663
Option 4 ID : 6306802760665
Status : Answered
Chosen Option : 1

Q.18 Select the option that expresses the given sentence in indirect speech.
He said to me, "I am very busy today."

- Ans
- ☐ 1. He said to me I am very busy today.
 - ☒ 2. He told me that he was very busy that day.
 - ☐ 3. He told me that he was very busy today.
 - ☐ 4. He said he is very today.

Question ID : 630680157380
Option 1 ID : 630680609053
Option 2 ID : 630680609055
Option 3 ID : 630680609052
Option 4 ID : 630680609054
Status : Answered
Chosen Option : 2



Q.19 Select the correct spelling of the underlined word in the given sentence.

The engineer's meticulus attention to detail ensured the structural integrity of the building.

Ans ☒ 1. meticolus

☒ 2. meticoulous

☒ 3. meticulous

☒ 4. mettculus

Question ID : 630680629326

Option 1 ID : 6306802463387

Option 2 ID : 6306802463386

Option 3 ID : 6306802463389

Option 4 ID : 6306802463388

Status : Answered

Chosen Option : 3

Q.20 The HCF and the LCM of two numbers are 22 and 308, respectively. If one of the numbers is $\frac{8}{7}$ times of the other, what is the value of the other number?

Ans ☒ 1. 88

☒ 2. 67

☒ 3. 113

☒ 4. 115

Question ID : 6306807

Option 1 ID : 6306803

Option 2 ID : 6306803

Option 3 ID : 6306803

Option 4 ID : 6306803

Status : Answered

Chosen Option : 1

Q.21 If 2 is added to each even and odd digit in the number 7245163, what will be the sum of the greatest and smallest digits in the new number thus formed?

Ans ☒ 1. 11

☒ 2. 14

☒ 3. 15

☒ 4. 12

Question ID : 630680372711

Option 1 ID : 6306801452132

Option 2 ID : 6306801452133

Option 3 ID : 6306801452131

Option 4 ID : 6306801452134

Status : Answered

Chosen Option : 4

Q.22 I, J, K, L, M and N live on six different floors of the same building. The bottommost floor of the building is numbered as 1; the floor above it is numbered as 2 and so on. The topmost floor is numbered as 6. M lives on the floor immediately below I. J lives on an even-numbered floor but not on floor number 6. Only two people live between M and K. L lives on the floor immediately below J but not on the bottommost floor. How many people live between M and N?

Ans ☒ 1. Two

☒ 2. One

☒ 3. Four

☒ 4. Three

Question ID : 630680333293

Option 1 ID : 6306801296446

Option 2 ID : 6306801296445

Option 3 ID : 6306801296448

Option 4 ID : 6306801296447

Status : Answered

Chosen Option : 4

Q.23 Select the option that expresses the given sentence in passive voice.
The coach had warned the team management of dire consequences if he was not paid on time.

Ans

✗ 1. The team management has been warned by the coach of dire consequences if he was not paid on time.

✗ 2. The team management is being warned by the coach of dire consequences if he was not paid on time.

✓ 3. The team management had been warned by the coach of dire consequences if he was not paid on time.

✗ 4. The team management was warned by the coach of dire consequences if he was not paid on time.

Question ID : 630680156612

Option 1 ID : 630680606106

Option 2 ID : 630680606104

Option 3 ID : 630680606107

Option 4 ID : 630680606105

Status : Answered

Chosen Option : 3

Q.24 Your colleagues just successfully solved a complex problem at work. Which of the following idioms would you use to congratulate them?

Ans

✗ 1. "Let's not jump the gun!"

✓ 2. "You hit the nail on the head!"

✗ 3. "Don't cry over spilled milk!"

✗ 4. "Burn the midnight oil!"

Question ID : 630680377265

Option 1 ID : 6306801470126

Option 2 ID : 6306801470125

Option 3 ID : 6306801470123

Option 4 ID : 6306801470124

Status : Answered

Chosen Option : 2

Q.25 In a certain code language, 'MAKE' is coded as '5210' and 'HAME' is coded as '1023'. What is the code for 'H' in the given code language?

Ans

✓ 1. 3

✗ 2. 2

✗ 3. 0

✗ 4. 1

Question ID : 630680988841

Option 1 ID : 6306803879557

Option 2 ID : 6306803879556

Option 3 ID : 6306803879555

Option 4 ID : 6306803879554

Status : Answered

Chosen Option : 1

Q.26 A man buys a second-hand car for ₹579000 and spends ₹10000 on its repairs. What should be its selling price (in ₹) if he wants to earn a 8% profit?

Ans

✗ 1. ₹628300

✓ 2. ₹636120

✗ 3. ₹633400

✗ 4. ₹639500

Question ID : 630680790040

Option 1 ID : 6306803095943

Option 2 ID : 6306803095940

Option 3 ID : 6306803095941

Option 4 ID : 6306803095942

Status : Answered

Chosen Option : 4

Q.27 In a certain code language, 'LAME' is coded as '5102' and 'MACE' is coded as '0235'. What is the code for 'C' in the given code language?

Ans

✗ 1. 2

✗ 2. 0

✗ 3. 5

✓ 4. 3

Question ID : 630680988838

Option 1 ID : 6306803879543

Option 2 ID : 6306803879542

Option 3 ID : 6306803879545

Option 4 ID : 6306803879544

Status : Answered

Chosen Option : 4

Q.28 Who among the following received the Arjuna Awards for outstanding performance in Sports and Games in 2023 in Archery?

- Ans ☒ 1. Ojas Pravin Deotale
☒ 2. Mohameed Hussamuddin
☒ 3. Krishan Bahadur Pathak
☒ 4. Sreeshankar M

Question ID : 6306801018168
Option 1 ID : 6306803996195
Option 2 ID : 6306803996197
Option 3 ID : 6306803996198
Option 4 ID : 6306803996196
Status : Answered
Chosen Option : 1

Q.29 A man spends 75% of his monthly salary (in ₹) on the rent of his house. If every month he also spends ₹945 on his conveyance and ₹1175 on his grocery and saves the remaining ₹296, his monthly salary (in ₹) is:

- Ans ☒ 1. 9664
☒ 2. 9730
☒ 3. 9590
☒ 4. 9762

Question ID : 630680789200
Option 1 ID : 6306803092580
Option 2 ID : 6306803092582
Option 3 ID : 6306803092581
Option 4 ID : 6306803092583
Status : Answered
Chosen Option : 1

Q.30 Sentences of a paragraph are given below in jumbled order. Arrange the sentences in the correct order to form a meaningful and coherent paragraph.

- A) The benefits of reading are manifold.
B) It improves concentration and expands knowledge.
C) Additionally, it enhances empathy and emotional intelligence.
D) Therefore, incorporating reading into your daily routine is highly recommended.

- Ans ☒ 1. ACBD
☒ 2. ADCB
☒ 3. ABCD
☒ 4. ABDC

Question ID : 630680993839
Option 1 ID : 6306803899456
Option 2 ID : 6306803899458
Option 3 ID : 6306803899455
Option 4 ID : 6306803899457
Status : Answered
Chosen Option : 1

Q.31 An open water drum is in the form of a cylinder. The height of the drum is 76 m and its internal diameter is 25 m. Find the area (in m², rounded off to one decimal place) that needs to be painted if one wants to paint the inside of the drum. Take $\pi = \frac{22}{7}$

- Ans ☒ 1. 6471.2
☒ 2. 6475.7
☒ 3. 6462.5
☒ 4. 6472.3

Question ID : 630680828183
Option 1 ID : 6306803244054
Option 2 ID : 6306803244053
Option 3 ID : 6306803244051
Option 4 ID : 6306803244052
Status : Not Answered
Chosen Option : --



Q.32 A goods train and a superfast train started running from station A and station B respectively at the same time towards each other on parallel tracks. If distance between station A and station B is 209 km and trains are running with speeds of 96 km/h and 113 km/h respectively, then at how much distance (in km) from station B will the trains meet?

- Ans
- ☒ 1. 133
 - ☒ 2. 127
 - ☒ 3. 113
 - ☒ 4. 121

Question ID : 630680894778
Option 1 ID : 6306803505132
Option 2 ID : 6306803505131
Option 3 ID : 6306803505129
Option 4 ID : 6306803505130
Status : Not Answered
Chosen Option : --

Q.33 What is the full form of ISO?

- Ans
- ☒ 1. Indian Standard Organisation
 - ☒ 2. International Standard Organization
 - ☒ 3. International Organization for Standardization
 - ☒ 4. Indian Organisation for Standardisation

Question ID : 630680983056
Option 1 ID : 6306803856188
Option 2 ID : 6306803856186
Option 3 ID : 6306803856187
Option 4 ID : 6306803856189
Status : Answered
Chosen Option : 1

Q.34 Select the pair which follows the same pattern as that followed by the two set of pairs given below. Both pairs follow the same pattern.

OTT : KVR
KHK : GJI

- Ans
- ☒ 1. APD : ERF
 - ☒ 2. TDD : PFB
 - ☒ 3. IQT : ESC
 - ☒ 4. BEP : FCN

Question ID : 630680449157
Option 1 ID : 6306801753787
Option 2 ID : 6306801753789
Option 3 ID : 6306801753788
Option 4 ID : 6306801753786
Status : Answered
Chosen Option : 2

Q.35 What should come in place of the question mark (?) in the given series?

901 902 ? 907 911 916

- Ans
- ☒ 1. 906
 - ☒ 2. 905
 - ☒ 3. 903
 - ☒ 4. 904

Question ID : 630680546539
Option 1 ID : 6306802136731
Option 2 ID : 6306802136730
Option 3 ID : 6306802136728
Option 4 ID : 6306802136729
Status : Answered
Chosen Option : 4

Ans  1. 40
 2. 70
 3. 30
 4. 60

Question ID : 630680375023
Option 1 ID : 6306801461260
Option 2 ID : 6306801461262
Option 3 ID : 6306801461259
Option 4 ID : 6306801461261
Status : Answered
Chosen Option : 3

△	K	□	△	○	□	◐	○	?
B	∞	∞	K	K	△	△	□	

1.  

2.  

3.  

4.  

Question ID : 630680564241
Option 1 ID : 6306802205919
Option 2 ID : 6306802205920
Option 3 ID : 6306802205922
Option 4 ID : 6306802205921
Status : Answered
Chosen Option : 4

Ans  1. 3
 2. 1
 3. 2
 4. 4

Question ID : 630680658426
Option 1 ID : 6306802577649
Option 2 ID : 6306802577646
Option 3 ID : 6306802577648
Option 4 ID : 6306802577647
Status : Answered
Chosen Option : 2

Q.39 G is the centroid of $\triangle ABC$. The medians AD and BE intersect at right angles. If the lengths of AD and BE are 7.5 cm and 18 cm, respectively, then the length of AB is:

- Ans ☒ 1. 13 cm
☒ 2. 14 cm
☒ 3. 10 cm
☒ 4. 11 cm

Question ID : 630680194653
 Option 1 ID : 630680755140
 Option 2 ID : 630680755137
 Option 3 ID : 630680755139
 Option 4 ID : 630680755138
 Status : Not Answered
 Chosen Option : --

Q.40 The theme of International Day of Happiness 2023 was _____.

- Ans ☒ 1. Happier Together
☒ 2. Keep Calm and Stay Happy
☒ 3. Be Happy, Be You
☒ 4. Be Mindful, Be Grateful, Be Kind

Question ID : 6306801056959
 Option 1 ID : 6306804150264
 Option 2 ID : 6306804150261
 Option 3 ID : 6306804150262
 Option 4 ID : 6306804150263
 Status : Answered
 Chosen Option : 4

Q.41 The area (in cm^2) of an equilateral triangle of side 49 cm is:

- Ans ☒ 1. $2401\sqrt{\frac{3}{4}}$
☒ 2. $2401\sqrt{\frac{3}{2}}$
☒ 3. $2401\sqrt{\frac{3}{3}}$
☒ 4. $2401\sqrt{\frac{3}{6}}$

Question ID : 6306801063323
 Option 1 ID : 6306804175655
 Option 2 ID : 6306804175657
 Option 3 ID : 6306804175656
 Option 4 ID : 6306804175658
 Status : Answered
 Chosen Option : 1

Q.42 Who among the following had written the novel 'The Dark Room'?

- Ans ☒ 1. RK Narayan
☒ 2. Mulk Raj Anand
☒ 3. Khushwant Singh
☒ 4. Ruskin Bond

Question ID : 630680404179
 Option 1 ID : 6306801576222
 Option 2 ID : 6306801576223
 Option 3 ID : 6306801576224
 Option 4 ID : 6306801576225
 Status : Answered
 Chosen Option : 4



Having a notably high temperature

Ans ☒ 1. Putative
☒ 2. Wary
☒ 3. Torrid
☒ 4. Assiduous

Question ID : 630680994118
Option 1 ID : 6306803900550
Option 2 ID : 6306803900549
Option 3 ID : 6306803900552
Option 4 ID : 6306803900551
Status : Answered
Chosen Option : 3

Ans ☒ 1. 13
☒ 2. 4
☒ 3. 15
☒ 4. 60

Question ID : 630680656657
Option 1 ID : 6306802570573
Option 2 ID : 6306802570572
Option 3 ID : 6306802570571
Option 4 ID : 6306802570570
Status : Answered
Chosen Option : 4

Ans  1. Prahasta

 2. Yukti

 3. Sadhanam

 4. Mangalam

Question ID : 630680992594
Option 1 ID : 6306803894650
Option 2 ID : 6306803894649
Option 3 ID : 6306803894648
Option 4 ID : 6306803894651
Status : Answered
Chosen Option : 2

Ans  1. 1929

 2. 1911

 3. 1890

 4. 1949

Question ID : 630680620640
Option 1 ID : 6306802428669
Option 2 ID : 6306802428670
Option 3 ID : 6306802428668
Option 4 ID : 6306802428667
Status : Answered
Chosen Option : 2

Ans  1. Securities and Exchange Board of India (SEBI)
 2. National Bank for Agriculture and Rural Development (NABARD)
 3. Industrial Finance Corporation of India (IFCI)
 4. Reserve Bank of India (RBI)

Question ID : 6306801056983
Option 1 ID : 6306804150358
Option 2 ID : 6306804150359
Option 3 ID : 6306804150360
Option 4 ID : 6306804150357
Status : Answered
Chosen Option : 2

Q.48 The monument 'Victoria Memorial' is situated in which Indian city?

- Ans ☒ 1. Chennai
☒ 2. Kolkata
☒ 3. Mumbai
☒ 4. Delhi

Question ID : 630680730201
Option 1 ID : 6306802861457
Option 2 ID : 6306802861455
Option 3 ID : 6306802861456
Option 4 ID : 6306802861458
Status : Answered
Chosen Option : 2

Q.49 Select the set in which the numbers are related in the same way as are the numbers of the following sets.

(NOTE: Operations should be performed on the whole numbers, without breaking down the numbers into its constituent digits. E.g. 13 – Operations on 13 such as adding/subtracting/multiplying etc. to 13 can be performed. Breaking down 13 into 1 and 3 and then performing mathematical operations on 1 and 3 is not allowed.)
(56, 112, 168)
(78, 156, 234)

- Ans ☒ 1. (64, 128, 192)
☒ 2. (76, 162, 218)
☒ 3. (73, 136, 219)
☒ 4. (51, 122, 163)

Question ID : 630680361336
Option 1 ID : 6306801407192
Option 2 ID : 6306801407191
Option 3 ID : 6306801407190
Option 4 ID : 6306801407189
Status : Answered
Chosen Option : 1

Q.50 Which of the following is an indicator used to analyse the employment situation in India?

- Ans ☒ 1. Head - count ratio
☒ 2. Employer-population ratio
☒ 3. Labour- population ratio
☒ 4. Worker- Population ratio

Question ID : 63068087403
Option 1 ID : 630680339004
Option 2 ID : 630680339001
Option 3 ID : 630680339002
Option 4 ID : 630680339003
Status : Answered
Chosen Option : 3

Q.51 If 'A' stands for '+', 'B' stands for 'x', 'C' stands for '+' and 'D' stands for '-', what will come in place of the question mark '?' in the following equation?
18 D 16 A 8 C 3 B 2 = ?

- Ans ☒ 1. 26
☒ 2. 22
☒ 3. 18
☒ 4. 12

Question ID : 630680374907
Option 1 ID : 6306801460798
Option 2 ID : 6306801460796
Option 3 ID : 6306801460797
Option 4 ID : 6306801460795
Status : Answered
Chosen Option : 2



Q.52 A grocery shop is started with a certain sum and gains annually at the rate of 30% compound interest. At the end of 3 years he has ₹10,985. What was his original sum?

- Ans
- ☒ 1. ₹6,500
 - ☒ 2. ₹6,000
 - ☒ 3. ₹5,000
 - ☒ 4. ₹5,500

Question ID : 630680524209
 Option 1 ID : 6306802048988
 Option 2 ID : 6306802048987
 Option 3 ID : 6306802048989
 Option 4 ID : 6306802048990
 Status : Answered
 Chosen Option : 3

Q.53 Select the most appropriate option to fill in the blank.

In most of _____ cases, it cannot be said for sure if it was murder or a suicide.

- Ans
- ☒ 1. No article required
 - ☒ 2. a
 - ☒ 3. the
 - ☒ 4. an

Question ID : 630680997134
 Option 1 ID : 6306803912247
 Option 2 ID : 6306803912244
 Option 3 ID : 6306803912246
 Option 4 ID : 6306803912245
 Status : Answered
 Chosen Option : 3

Q.54 Two numbers are in the ratio 3 : 8. If the first number is increased by 9 and the second number is decreased by 3, then the ratio becomes 6 : 7. What is the sum of the original two numbers?

- Ans
- ☒ 1. 33
 - ☒ 2. 32
 - ☒ 3. 30
 - ☒ 4. 31

Question ID : 630680954638
 Option 1 ID : 6306803742760
 Option 2 ID : 6306803742763
 Option 3 ID : 6306803742761
 Option 4 ID : 6306803742762
 Status : Answered
 Chosen Option : 1

Q.55 Seven people P, Q, R, S, T, U and V are sitting in a straight row, facing north. B sits second to the left of F. No one sits to the right of F. Only three people sit between A and D. A is not an immediate neighbor of F. Only two people sit between G and E. E is not an immediate neighbor of A. Who sits at the extreme left?

- Ans
- ☒ 1. E
 - ☒ 2. A
 - ☒ 3. C
 - ☒ 4. G

Question ID : 630680520875
 Option 1 ID : 6306802035925
 Option 2 ID : 6306802035926
 Option 3 ID : 6306802035928
 Option 4 ID : 6306802035927
 Status : Answered
 Chosen Option : 4

Comprehension:

Read the given passage and answer the questions that follow.

When you think about housework, you likely think of actions: scrubbing the dishes, running errands, chopping vegetables for dinner. And it's not news to say that mothers often shoulder the burden for most of those activities. But there's an invisible dimension of household labour that unfolds behind the scenes: the cognitive effort that goes into anticipating needs, planning, organising and delegating household tasks. In other words, someone has to remember to replace the dish soap and select which vegetables to chop.

Our new research found that this cognitive dimension of housework, often called the 'mental load,' is divided even more unevenly within couples than the physical dimension – and it seems to take a particular mental health toll on women. According to the study we published in the Archives of Women's Mental Health, mothers who take on a more disproportionate share of cognitive labour report higher levels of depression, stress, relationship dissatisfaction and burnout. An unequal division of household labour is a key driver of global gender inequity, suppressing women's full participation in the paid workforce and significantly affecting women's health and well-being.

Our study is one of the first to investigate the cognitive dimension of household labour and its effects on maternal mental health. Cognitive labour may be particularly taxing for women because it often runs behind the scenes and goes unacknowledged or unappreciated by others. It also pulls mental energy away from other priorities. Additional studies indicate that women experience more negative effects from childcare and housework compared to men, such as higher depression rates, partly due to the heavier cognitive load they carry.

SubQuestion No : 56

Q.56 What can be inferred about the impact of cognitive labour on gender equity?

- Ans**
- ☒ 1. Cognitive labour is equally divided and does not affect gender equity.
 - ☒ 2. Cognitive labour contributes to gender equity by promoting equal participation in housework.
 - ☒ 3. Cognitive labour has no significant impact on gender equity.
 - ☒ 4. Cognitive labour perpetuates gender inequity by suppressing women's participation in the paid workforce.

Question ID : 630680998437
Option 1 ID : 6306803917637
Option 2 ID : 6306803917635
Option 3 ID : 6306803917634
Option 4 ID : 6306803917636
Status : Answered
Chosen Option : 2

Comprehension:

Read the given passage and answer the questions that follow.

When you think about housework, you likely think of actions: scrubbing the dishes, running errands, chopping vegetables for dinner. And it's not news to say that mothers often shoulder the burden for most of those activities. But there's an invisible dimension of household labour that unfolds behind the scenes: the cognitive effort that goes into anticipating needs, planning, organising and delegating household tasks. In other words, someone has to remember to replace the dish soap and select which vegetables to chop.

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SubQuestion No : 57

Q.57 According to the passage, what is 'mental load'?

- Ans**
- ☒ 1. The financial cost of running a household
 - ☒ 2. The burden of childcare activities
 - ☒ 3. The physical effort required for housework
 - ☒ 4. The cognitive effort involved in managing household tasks

Question ID : 630680998436
Option 1 ID : 6306803917633
Option 2 ID : 6306803917632
Option 3 ID : 6306803917630
Option 4 ID : 6306803917631
Status : Answered
Chosen Option : 4

Comprehension:

Read the given passage and answer the questions that follow.

When you think about housework, you likely think of actions: scrubbing the dishes, running errands, chopping vegetables for dinner. And it's not news to say that mothers often shoulder the burden for most of those activities. But there's an invisible dimension of household labour that unfolds behind the scenes: the cognitive effort that goes into anticipating needs, planning, organising and delegating household tasks. In other words, someone has to remember to replace the dish soap and select which vegetables to chop.

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SubQuestion No : 58

Q.58 What is the tone of the passage?

- Ans ☒ 1. Optimistic and encouraging
- ☒ 2. Sympathetic and analytical
- ☒ 3. Critical and dismissive
- ☒ 4. Neutral and detached

Question ID : 630680998439
Option 1 ID : 6306803917644
Option 2 ID : 6306803917643
Option 3 ID : 6306803917645
Option 4 ID : 6306803917642
Status : Answered
Chosen Option : 2

Comprehension:

Read the given passage and answer the questions that follow.

When you think about housework, you likely think of actions: scrubbing the dishes, running errands, chopping vegetables for dinner. And it's not news to say that mothers often shoulder the burden for most of those activities. But there's an invisible dimension of household labour that unfolds behind the scenes: the cognitive effort that goes into anticipating needs, planning, organising and delegating household tasks. In other words, someone has to remember to replace the dish soap and select which vegetables to chop.

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SubQuestion No : 59

Q.59 How is the passage structured to present its argument?

- Ans ☒ 1. Presentation of study findings, discussion of physical and cognitive labour, conclusion on gender roles and mental health
- ☒ 2. Discussion of physical housework, presentation of gender roles, introduction to cognitive labour, conclusion on mental health
- ☒ 3. Introduction to housework actions, discussion of cognitive labour, presentation of study findings, conclusion on gender equity
- ☒ 4. Overview of gender inequity, analysis of housework impact, presentation of solutions, discussion of future research

Question ID : 630680998438
Option 1 ID : 6306803917641
Option 2 ID : 6306803917639
Option 3 ID : 6306803917638
Option 4 ID : 6306803917640
Status : Answered
Chosen Option : 2

Comprehension:

Read the given passage and answer the questions that follow.

When you think about housework, you likely think of actions: scrubbing the dishes, running errands, chopping vegetables for dinner. And it's not news to say that mothers often shoulder the burden for most of those activities. But there's an invisible dimension of household labour that unfolds behind the scenes: the cognitive effort that goes into anticipating needs, planning, organising and delegating household tasks. In other words, someone has to remember to replace the dish soap and select which vegetables to chop.

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SubQuestion No : 60

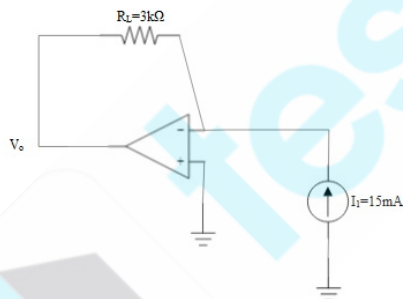
Q.60 Which of the following real-life scenarios best connects to the issues discussed in the passage?

- Ans
- ✓ 1. A support group for mothers experiencing burnout due to household responsibilities
 - ✗ 2. A campaign advocating for equal pay for women in the workforce
 - ✗ 3. A workshop on effective physical fitness routines for busy parents
 - ✗ 4. A community event promoting local organic vegetable markets

Question ID : 630680998440
Option 1 ID : 6306803917649
Option 2 ID : 6306803917646
Option 3 ID : 6306803917647
Option 4 ID : 6306803917648
Status : Answered
Chosen Option : 1

Section : Knowledge of Discipline

Q.1 Given is a trans-resistance amplifier circuit. What would be the value of the output voltage?



- Ans
- ✓ 1. - 45 V
 - ✗ 2. 45 V
 - ✗ 3. - 15 V
 - ✗ 4. 15 V

Question ID : 630680117053
Option 1 ID : 630680453498
Option 2 ID : 630680453499
Option 3 ID : 630680453501
Option 4 ID : 630680453500
Status : Answered
Chosen Option : 1

Q.2 The capacitance of a single isolated sphere of radius 3cm is ____.

- Ans
- ✓ 1. 3.33pF
 - ✗ 2. 2.67pF
 - ✗ 3. 4.567pF
 - ✗ 4. 6.543pF

Question ID : 63068063814
Option 1 ID : 630680246347
Option 2 ID : 630680246346
Option 3 ID : 630680246348
Option 4 ID : 630680246349
Status : Answered
Chosen Option : 3

Q.3 The standard second order system whose characteristic equation is given by
 $P(s) = a_2s^2 + a_1s + a_0$
is stable if:

- Ans
- ✗ 1. even power of 's' has negative coefficients and odd power have positive coefficients
 - ✓ 2. all the coefficients are positive or if all the coefficients are negative
 - ✗ 3. all coefficients must be strictly positive
 - ✗ 4. even power of 's' has positive coefficients and odd power have negative coefficients

Question ID : 63068093722
Option 1 ID : 630680363585
Option 2 ID : 630680363584
Option 3 ID : 630680363586
Option 4 ID : 630680363583
Status : Answered
Chosen Option : 1

Q.4 Form the differential equation by eliminating the arbitrary constants from $y = Ae^{3x} + Be^{2x}$.

- Ans
- ✓ 1. $\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0$
 - ✗ 2. $\frac{d^2y}{dx^2} + \frac{dy}{dx} + 6y = 0$
 - ✗ 3. $5\frac{d^2y}{dx^2} + \frac{dy}{dx} - 6y = 0$
 - ✗ 4. $\frac{d^2y}{dx^2} + \frac{dy}{5dx} - 6y = 0$

Question ID : 63068064964
Option 1 ID : 630680250824
Option 2 ID : 630680250822
Option 3 ID : 630680250825
Option 4 ID : 630680250823
Status : Not Answered
Chosen Option : --

Q.5 Angular speed of separately excited DC motor is given by:

- Ans
- ✗ 1. $\omega_m = \frac{V_t - I_a^2 r_a}{K_a \phi}$
 - ✗ 2. $\omega_m = \frac{V_t - I_a(r_a + r_s)}{k}$
 - ✓ 3. $\omega_m = \frac{V_t - I_a r_a}{K_a \phi}$
 - ✗ 4. $\omega_m = \frac{V_t - I_a(r_a + r_s)}{k I_a}$

Question ID : 63068064427
Option 1 ID : 630680248694
Option 2 ID : 630680248697
Option 3 ID : 630680248695
Option 4 ID : 630680248696
Status : Answered
Chosen Option : 4



Q.6 The quality of measurement is given by _____.

- Ans
- ☒ 1. true value
 - ☒ 2. error
 - ☒ 3. relative static error
 - ☒ 4. indicated value

Question ID : 63068094527
Option 1 ID : 630680366803
Option 2 ID : 630680366805
Option 3 ID : 630680366806
Option 4 ID : 630680366804
Status : Answered
Chosen Option : 2

Q.7 Starting the sequence from the number with the bar on top, the z-transform of the sequence {2, 5, 1, -5, 3} will be:

- Ans
- ☒ 1. $X(z) = -2 - 5z^1 - z^2 + 5z^3 - 3z^4$
 - ☒ 2. $X(z) = 2 + 5z^1 + z^2 - 5z^3 + 3z^4$
 - ☒ 3. $X(z) = -2 - 5z^1 - z^2 + 5z^3 - 3z^4$
 - ☒ 4. $X(z) = 2 + 5z^1 + z^2 - 5z^3 + 3z^4$

Question ID : 630680203465
Option 1 ID : 630680789062
Option 2 ID : 630680789061
Option 3 ID : 630680789060
Option 4 ID : 630680789059
Status : Not Answered
Chosen Option : --

Q.8 Which of the following statements is true for FIR filters?

- Ans
- ☒ 1. FIR can realise frequency response with a linear phase.
 - ☒ 2. FIR digital filters have infinite memory.
 - ☒ 3. FIR can realise with a shorter filter length.
 - ☒ 4. FIR can realise frequency response with a non-linear phase.

Question ID : 630680764600
Option 1 ID : 6306802996921
Option 2 ID : 6306802996919
Option 3 ID : 6306802996920
Option 4 ID : 6306802996922
Status : Answered
Chosen Option : 4

Q.9 The discrete time unit impulse is the _____ of the discrete time step signal.

- Ans
- ☒ 1. first difference
 - ☒ 2. running sum
 - ☒ 3. running integral
 - ☒ 4. running derivative

Question ID : 630680734802
Option 1 ID : 6306802879988
Option 2 ID : 6306802879986
Option 3 ID : 6306802879985
Option 4 ID : 6306802879987
Status : Answered
Chosen Option : 3

Q.10 Loading effect is primarily caused by instruments having:

- Ans
- ☒ 1. low sensitivity
 - ☒ 2. high sensitivity
 - ☒ 3. high range
 - ☒ 4. high resistance

Question ID : 630680104231
Option 1 ID : 630680405141
Option 2 ID : 630680405140
Option 3 ID : 630680405142
Option 4 ID : 630680405139
Status : Answered
Chosen Option : 1



Q.11 What is the relation between a square matrix A, its eigenvalue λ and eigenvector v ?

Ans

✗ 1. $A.A = \lambda v$

✓ 2. $Av = \lambda v$

✗ 3. $\lambda A = . v.v$

✗ 4. No universal relation exists

Question ID : 630680103904

Option 1 ID : 630680403834

Option 2 ID : 630680403832

Option 3 ID : 630680403833

Option 4 ID : 630680403831

Status : Answered

Chosen Option : 2

Q.12 Consider the polar plot shown in the following figure. $G(j\omega) H(j\omega)$ is open-loop gain.

Ans

✗ 1. unstable and has three closed-loop poles in the right half of S plane

✗ 2. unstable and has a single closed-loop pole in the right half of S plane

✓ 3. stable, as there is no closed-loop pole in the right half of S plane

✗ 4. unstable and has two closed-loop poles in the right half of S plane

Question ID : 630680402891

Option 1 ID : 6306801571159

Option 2 ID : 6306801571157

Option 3 ID : 6306801571160

Option 4 ID : 6306801571158

Status : Answered

Chosen Option : 2

Q.13 Let $A = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$. Express $4A^5 - 16A^4 - 28A^3 + 28A^2 + 60A + 30I$ as a linear polynomial.

Ans

✓ 1. $4A + 10I$

✗ 2. $10A - 4I$

✗ 3. $4A - 10I$

✗ 4. $10A + 4I$

Question ID : 630680103275

Option 1 ID : 630680401359

Option 2 ID : 630680401361

Option 3 ID : 630680401360

Option 4 ID : 630680401362

Status : Not Answered

Chosen Option : --

Q.14 Which of the following statements best describes a significant advantage of using Karnaugh Maps (K-maps) over algebraic simplification in the context of digital logic design?

Ans

✗ 1. K-maps automatically minimise the number of product terms without the need for Boolean algebraic manipulation, thus ensuring fewer terms compared to algebraic simplification.

✓ 2. K-maps offer a visual method for grouping terms, which can simplify the Boolean expressions more effectively and intuitively than algebraic methods, especially for functions with more than four variables.

✗ 3. K-maps always provide a minimal expression with fewer literals than the algebraic method.

✗ 4. K-maps can be used to directly implement the Boolean function into a hardware circuit without requiring any further simplification or logic gate minimisation.

Question ID : 6306801038551

Option 1 ID : 6306804076861

Option 2 ID : 6306804076860

Option 3 ID : 6306804076859

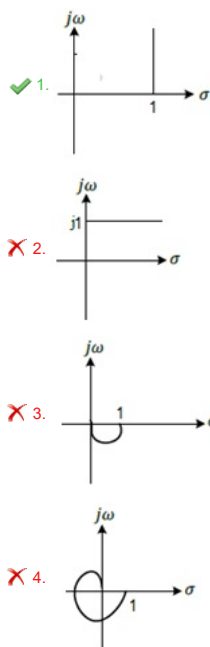
Option 4 ID : 6306804076862

Status : Answered

Chosen Option : 2

Q.15 For the transfer function $(j\omega) = 1 + j\omega$, the corresponding polar plot for all ranges of positive frequency resembles to:

Ans



Question ID : 630680402890
Option 1 ID : 6306801571154
Option 2 ID : 6306801571155
Option 3 ID : 6306801571153
Option 4 ID : 6306801571156
Status : Answered
Chosen Option : 3

Q.16 In a linear, isotropic medium, the relationship between the magnetic field intensity $\vec{H}$ and the magnetic flux density $\vec{B}$ is described by:

- Ans
1. $\vec{B} = \mu_0 \vec{H}$
2. $\vec{B} = \frac{\vec{H}}{\mu_0 \mu_r}$
3. $\vec{B} = \frac{\vec{H}}{\mu_0}$
4. $\vec{B} = \mu_0 \mu_r \vec{H}$

Question ID : 6306801038197
Option 1 ID : 6306804075576
Option 2 ID : 6306804075578
Option 3 ID : 6306804075575
Option 4 ID : 6306804075577
Status : Answered
Chosen Option : 1

Q.17 Which of the following components in a steam power station is responsible for increasing the temperature of air before it enters the boiler furnace, thereby improving the thermal efficiency of the system?

- Ans
1. Economiser
2. Condenser
3. Superheater
4. Air Preheater

Question ID : 6306801053195
Option 1 ID : 6306804135371
Option 2 ID : 6306804135373
Option 3 ID : 6306804135370
Option 4 ID : 6306804135372
Status : Answered
Chosen Option : 1



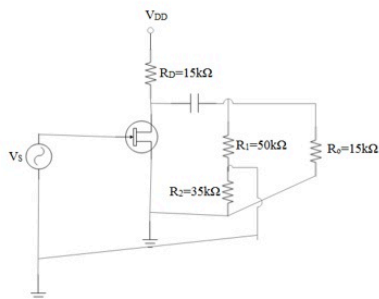
Q.18 The principle of the Carey Foster bridge is similar to that of:

- Ans
- ☒ 1. Hay's bridge
 - ☒ 2. Campbell's bridge
 - ☒ 3. Campbell's modification of Heaviside bridge
 - ☒ 4. Wheatstone bridge

Question ID : 630680133513
 Option 1 ID : 630680517027
 Option 2 ID : 630680517028
 Option 3 ID : 630680517029
 Option 4 ID : 630680517026
 Status : Answered
 Chosen Option : 3

Q.19 What would be the gain of the FET amplifier circuit without feedback? Given

$g_m = 3500 \mu S$.



- Ans
- ☒ 1. -25.86
 - ☒ 2. -25.27
 - ☒ 3. -26.25
 - ☒ 4. -26.78

Question ID : 630680117038
 Option 1 ID : 630680453438
 Option 2 ID : 630680453440
 Option 3 ID : 630680453439
 Option 4 ID : 630680453441
 Status : Not Answered
 Chosen Option : --

Q.20 In the study of electromagnetic waves, the average Poynting vector (S) represents the

- Ans
- ☒ 1. instantaneous power per unit volume
 - ☒ 2. time-averaged electric field energy
 - ☒ 3. average power per unit area flowing through a surface
 - ☒ 4. peak electric field strength



Question ID : 6306801037752
 Option 1 ID : 6306804073837
 Option 2 ID : 6306804073840
 Option 3 ID : 6306804073838
 Option 4 ID : 6306804073839
 Status : Answered
 Chosen Option : 3

Q.21 Which of the following statements is true about paramagnetic susceptibility?

- Ans
- ☒ 1. It is independent of temperature.
 - ☒ 2. It varies inversely with absolute temperature.
 - ☒ 3. It varies directly with absolute temperature.
 - ☒ 4. It remains constant regardless of the magnetic field strength.

Question ID : 6306801043601
 Option 1 ID : 6306804096552
 Option 2 ID : 6306804096553
 Option 3 ID : 6306804096554
 Option 4 ID : 6306804096555
 Status : Answered
 Chosen Option : 2

Q.22 Which of the following statements about antilog amplifier is correct?

Ans ☒ 1. Due to infinite output impedance of op-amp the current flowing through the diode flows opposite to the feedback path through the resistor.

☒ 2. Due to infinite input impedance of op-amp the current flowing through the diode flows opposite the feedback path through the resistor.

☒ 3. Due to infinite input impedance of op-amp the current flowing through the diode flows along the feedback path through the resistor.

☒ 4. Due to infinite output impedance of op-amp the current flowing through the diode flows along the feedback path through the resistor.

Question ID : 630680117003

Option 1 ID : 630680453299

Option 2 ID : 630680453301

Option 3 ID : 630680453300

Option 4 ID : 630680453298

Status : Answered

Chosen Option : 1

Q.23 _____ is the excess-3 code for 123.

Ans ☒ 1. 010110010111

☒ 2. 010001010110

☒ 3. 010110101101

☒ 4. 100010100111

Question ID : 630680103559

Option 1 ID : 630680402481

Option 2 ID : 630680402479

Option 3 ID : 630680402482

Option 4 ID : 630680402480

Status : Answered

Chosen Option : 4

Q.24 In an axial blast circuit breaker, what happens when a tripping impulse opens the air valve?

Ans ☒ 1. The arc is moved to the fixed contact.

☒ 2. The air reservoir tank fills with air.

☒ 3. The air pressure is released, causing the circuit breaker to open.

☒ 4. The air exerts pressure on the moving contacts, causing them to move.

Question ID : 6306801053251

Option 1 ID : 6306804135596

Option 2 ID : 6306804135595

Option 3 ID : 6306804135594

Option 4 ID : 6306804135597

Status : Answered

Chosen Option : 4

Q.25 In Class A Chopper, The output voltage is zero during OFF condition but output current continuously flow due to

Ans ☒ 1. applied voltage

☒ 2. short circuit condition

☒ 3. capacitor

☒ 4. freewheeling diode

Question ID : 63068093444

Option 1 ID : 630680362471

Option 2 ID : 630680362474

Option 3 ID : 630680362472

Option 4 ID : 630680362473

Status : Answered

Chosen Option : 4

Q.26 The equivalent capacitance of two capacitors of 14F and 20F capacitance, when connected in parallel is_____.

Ans ☒ 1. 6F

☒ 2. 34F

☒ 3. 24F

☒ 4. 12F

Question ID : 63068063807

Option 1 ID : 630680246321

Option 2 ID : 630680246320

Option 3 ID : 630680246318

Option 4 ID : 630680246319

Status : Answered

Chosen Option : 2

Q.27 Cycle stealing allows Direct Memory Access (DMA) Controller to transfer _____ data word(s) at a time.

- Ans
- ☒ 1. 4
 - ☒ 2. 3
 - ☒ 3. 1
 - ☒ 4. 2

Question ID : 63068094475
Option 1 ID : 630680366598
Option 2 ID : 630680366597
Option 3 ID : 630680366595
Option 4 ID : 630680366596
Status : Answered
Chosen Option : 4

Q.28 In the transistor present in a Hartley oscillator, which of the following feedback circuits are used?

- Ans
- ☒ 1. Inductive fixed bias
 - ☒ 2. Capacitive fixed bias
 - ☒ 3. Capacitive voltage divider
 - ☒ 4. Inductive voltage divider

Question ID : 630680119941
Option 1 ID : 630680464830
Option 2 ID : 630680464831
Option 3 ID : 630680464833
Option 4 ID : 630680464832
Status : Answered
Chosen Option : 4

Q.29 How is slate made non-hygroscopic for use in electrical applications?

- Ans
- ☒ 1. By coating with insulating varnish
 - ☒ 2. By carving it into specific shapes
 - ☒ 3. By increasing its thickness
 - ☒ 4. By polishing

Question ID : 6306801041430
Option 1 ID : 6306804087988
Option 2 ID : 6306804087990
Option 3 ID : 6306804087989
Option 4 ID : 6306804087987
Status : Answered
Chosen Option : 2

Q.30 Consider a narrow band FM signal approximately defined by $s(t) \cong A_c \cos(2\pi f_c t) - \beta A_c \sin(2\pi f_c t) \sin(2\pi f_m t)$.

If P_t is the average power of $s(t)$ and P_c is the average carrier power, the P_t/P_c is:

- Ans
- ☒ 1. $\sqrt{1 + \beta^2}$
 - ☒ 2. $\sqrt{1 + \frac{\beta^2}{2}}$
 - ☒ 3. $1 + \frac{\beta^2}{2}$
 - ☒ 4. $1 + \beta^2$

Question ID : 630680146379
Option 1 ID : 630680566400
Option 2 ID : 630680566399
Option 3 ID : 630680566398
Option 4 ID : 630680566401
Status : Answered
Chosen Option : 2

Q.31 The empirical relation for bandwidth of a frequency modulated wave is given by:

- Ans
- ☒ 1. Fleming's right-hand rule
 - ☒ 2. Carson's rule
 - ☒ 3. Fleming's left-hand rule
 - ☒ 4. Euler's formula

Question ID : 630680744214
Option 1 ID : 6306802917087
Option 2 ID : 6306802917089
Option 3 ID : 6306802917086
Option 4 ID : 6306802917088
Status : Answered
Chosen Option : 2

Q.32 What is a key benefit of using nanostructured materials in dye-sensitised solar cells?

- Ans
- ☒ 1. Enhanced efficiency in converting solar energy to electricity
 - ☒ 2. Reduced production cost
 - ☒ 3. Increased thermal resistance
 - ☒ 4. Extended lifespan of the solar cell

Question ID : 6306801046147
Option 1 ID : 6306804106674
Option 2 ID : 6306804106672
Option 3 ID : 6306804106673
Option 4 ID : 6306804106675
Status : Answered
Chosen Option : 1

Q.33 What is the z-transform and ROC of the signal $[a^n \cos \omega_0 n]u[n]$?

- Ans
- ☒ 1. $\frac{1 - [a \cos \omega_0]z^{-1}}{1 + [2a \cos \omega_0]z^{-1} - a^2 z^{-2}}, |z| < a$
 - ☒ 2. $\frac{1 - [a \cos \omega_0]z^{-1}}{1 - [2a \cos \omega_0]z^{-1} + a^2 z^{-2}}, |z| < a$
 - ☒ 3. $\frac{1 - [a \cos \omega_0]z^{-1}}{1 + [2a \cos \omega_0]z^{-1} - a^2 z^{-2}}, |z| > a$
 - ☒ 4. $\frac{1 - [a \cos \omega_0]z^{-1}}{1 - [2a \cos \omega_0]z^{-1} + a^2 z^{-2}}, |z| > a$

Question ID : 630680203425
Option 1 ID : 630680788901
Option 2 ID : 63068078899
Option 3 ID : 630680788902
Option 4 ID : 630680788900
Status : Not Answered
Chosen Option : --

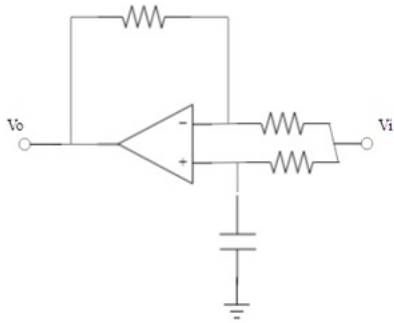
Q.34 Ampère's Circuital Law in its integral form states that:

- Ans
- ☒ 1. $\oint_c \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{enc}$
 - ☒ 2. $\oint_c \mathbf{B} \cdot d\mathbf{l} = \frac{1}{\mu_0} \frac{d\phi_E}{dt}$
 - ☒ 3. $\oint_c \mathbf{E} \cdot d\mathbf{l} = - \frac{d\phi_B}{dt}$
 - ☒ 4. $\oint_c \mathbf{B} \cdot d\mathbf{l} = \mu_0 \epsilon_0 \frac{d\phi_E}{dt}$

Question ID : 6306801038101
Option 1 ID : 6306804075197
Option 2 ID : 6306804075199
Option 3 ID : 6306804075198
Option 4 ID : 6306804075196
Status : Answered
Chosen Option : 1



Q.35 The given circuit represents which kind of filter?



- Ans ☒ 1. All pass filter
☐ 2. Band stop filter
☐ 3. Band pass filter
☐ 4. High pass filter

Question ID : 630680119932
Option 1 ID : 630680464796
Option 2 ID : 630680464797
Option 3 ID : 630680464795
Option 4 ID : 630680464794
Status : Answered
Chosen Option : 4

Q.36 The charge in a 5F capacitor connected to 100V source is_____.

- Ans ☒ 1. 500C
☐ 2. 100C
☐ 3. 300C
☐ 4. 50C

Question ID : 63068063809
Option 1 ID : 630680246329
Option 2 ID : 630680246327
Option 3 ID : 630680246328
Option 4 ID : 630680246326
Status : Answered
Chosen Option : 1

Q.37 A bank of single-phase transformers connected in delta/delta can be put in parallel with which of the following combinations of a three-phase connection?

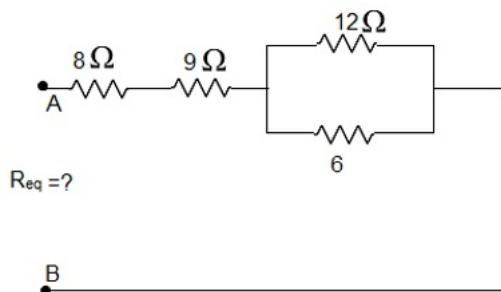
- 1) Star/Star
2) Star/Delta
3) Delta/Star
4) Delta/Delta

- Ans ☐ 1. 2 and 4
☐ 2. 1 and 2
☐ 3. 1 and 3
☒ 4. 1 and 4

Question ID : 63068061761
Option 1 ID : 630680238365
Option 2 ID : 630680238363
Option 3 ID : 630680238364
Option 4 ID : 630680238362
Status : Answered
Chosen Option : 2



Q.38 Study the given circuit. Calculate R_{eq} between A and B.



- Ans
- ✓ 1. 21 Ω
 - ✗ 2. 23 Ω
 - ✗ 3. 35 Ω
 - ✗ 4. 15 Ω

Question ID : 630680117092
 Option 1 ID : 630680453654
 Option 2 ID : 630680453656
 Option 3 ID : 630680453655
 Option 4 ID : 630680453657
 Status : Answered
 Chosen Option : 1

Q.39 The mathematical expression for the polarization P of a dielectric material is represented as _____.

- Ans
- ✗ 1. $P = \frac{D - \epsilon_0 E}{\epsilon_0}$
 - ✗ 2. $P = \epsilon_0 E \left(1 - \frac{1}{k}\right)$
 - ✓ 3. $P = k E$
 - ✗ 4. $P = \epsilon_0 E$

Question ID : 6306801038637
 Option 1 ID : 6306804077133
 Option 2 ID : 6306804077132
 Option 3 ID : 6306804077134
 Option 4 ID : 6306804077131
 Status : Answered
 Chosen Option : 2

Q.40 Why are magnetically soft materials NOT suitable for making permanent magnets?

- Ans
- ✗ 1. They have high coercive force.
 - ✓ 2. They have negligible coercive force.
 - ✗ 3. They store a lot of magnetic energy.
 - ✗ 4. They have high remanence.

Question ID : 6306801046126
 Option 1 ID : 6306804106597
 Option 2 ID : 6306804106598
 Option 3 ID : 6306804106599
 Option 4 ID : 6306804106596
 Status : Answered
 Chosen Option : 1

Q.41 What is the main objective of the economic load dispatch problem?

- Ans
- ✗ 1. To fix the generation at all generator buses
 - ✗ 2. To maximise the generation at the slack bus
 - ✗ 3. To reduce the operating reserve margin
 - ✓ 4. To meet load demand with minimum fuel consumption

Question ID : 6306801050163
 Option 1 ID : 6306804122997
 Option 2 ID : 6306804122999
 Option 3 ID : 6306804123000
 Option 4 ID : 6306804122998
 Status : Answered
 Chosen Option : 4

Q.42 Consider an angle modulated signal with carrier frequency 1 MHz, frequency deviation 10 KHz and message signal bandwidth as 5 KHz, is input to a multiplier of 20. The output signal carrier frequency and frequency deviation, respectively, are:

- Ans
- ☒ 1. 1 MHz and 200 KHz
 - ☒ 2. 20 MHz and 10 KHz
 - ☒ 3. 20 MHz and 20 KHz
 - ☒ 4. 20 MHz and 200 KHz

Question ID : 630680146382
Option 1 ID : 630680566413
Option 2 ID : 630680566411
Option 3 ID : 630680566412
Option 4 ID : 630680566410
Status : Answered
Chosen Option : 4

Q.43 Consider the Boolean function $f(A,B,C,D)$ with the minterms given by $\sum m(0, 1, 4, 5, 8, 9, 12, 13)$. Which of the following pairs of expressions determine the correct minimised forms for both the Sum of Products (SOP) and Product of Sums (POS) representations of the function?

- Ans
- ☒ 1. SOP: $A'B+A'B'$
POS: $(A+B+C)(A+B'+C)(A'+B+C)(A'+B'+C)$
 - ☒ 2. SOP: $A'C'+B'C'$
POS: $(C'+A+B)(C'+A'+B')$
 - ☒ 3. SOP: C'
POS: C'
 - ☒ 4. SOP: $A'B+B'C$
POS: $(A+C+D)(A'+B'+D)(B+C'+D')$

Question ID : 6306801038766
Option 1 ID : 6306804077479
Option 2 ID : 6306804077481
Option 3 ID : 6306804077480
Option 4 ID : 6306804077482
Status : Not Answered
Chosen Option : --

Q.44 _____ code is also known as reflected code.

- Ans
- ☒ 1. BCD code
 - ☒ 2. Gray code
 - ☒ 3. Binary code
 - ☒ 4. ASCII code

Question ID : 630680103577
Option 1 ID : 630680402551
Option 2 ID : 630680402554
Option 3 ID : 630680402552
Option 4 ID : 630680402553
Status : Answered
Chosen Option : 4

Q.45 If $x(n)$ is a right-sided sequence and if the circle $z = r_0$ is in the ROC, then all the finite values of z for which _____ will also be in the ROC.

- Ans
- ☒ 1. $|z| < r_0$
 - ☒ 2. $|z| > r_0$
 - ☒ 3. $0 < |z| < r_0$
 - ☒ 4. $|z| = r_0$

Question ID : 630680764603
Option 1 ID : 6306802996932
Option 2 ID : 6306802996931
Option 3 ID : 6306802996934
Option 4 ID : 6306802996933
Status : Not Answered
Chosen Option : --

Q.46 Which is the best suitable chopper for speed regulation and regenerative braking on a DC motor vehicle?

- Ans ☒ 1. B type
☒ 2. D type
☒ 3. A type
☒ 4. C type

Question ID : 630680185361
Option 1 ID : 630680718394
Option 2 ID : 630680718396
Option 3 ID : 630680718393
Option 4 ID : 630680718395
Status : Answered
Chosen Option : 4

Q.47 Which of the following best describes the current direction in compensating windings relative to the armature conductors?

- Ans ☒ 1. The current flows perpendicular to the armature conductors.
☒ 2. The current flows in the opposite direction to the armature conductors directly below the pole shoes.
☒ 3. The current flows in the same direction as the armature conductors.
☒ 4. The current alternates in direction depending on the load.

Question ID : 6306801050162
Option 1 ID : 6306804122995
Option 2 ID : 6306804122994
Option 3 ID : 6306804122993
Option 4 ID : 6306804122996
Status : Answered
Chosen Option : 2

Q.48 If both inputs of a half adder are high , then _____.

- Ans ☒ 1. S = 0, C = 1
☒ 2. S = 1, C = 0
☒ 3. S = 1, C = 1
☒ 4. S = 0, C = 0

Question ID : 630680103565
Option 1 ID : 630680402505
Option 2 ID : 630680402506
Option 3 ID : 630680402504
Option 4 ID : 630680402503
Status : Answered
Chosen Option : 4

Q.49 The range of 2's complement of binary number is (for n bit size) _____.

- Ans ☒ 1. -2^{n+1} to $+2^{n+1}$
☒ 2. -2^{n-1} to $+2^{n+1}$
☒ 3. -2^{n-1} to $+2^{n-1}-1$
☒ 4. -2^{n-1} to $+2^{n-1}$

Question ID : 630680103528
Option 1 ID : 630680402355
Option 2 ID : 630680402357
Option 3 ID : 630680402358
Option 4 ID : 630680402356
Status : Answered
Chosen Option : 4

Q.50 What is the range of dimensions for the nanoregime?

- Ans ☒ 1. 100 to 150 nm
☒ 2. 0.1 to 10 nm
☒ 3. 1 to 100 nm
☒ 4. Greater than 150 nm

Question ID : 6306801046159
Option 1 ID : 6306804106734
Option 2 ID : 6306804106732
Option 3 ID : 6306804106733
Option 4 ID : 6306804106735
Status : Answered
Chosen Option : 3



Q.51 Consider an RLC series circuit with circuit components R, L and C with the output across the resistor. The half power occurs for frequencies, which can be found by equating the magnitude of the transfer function to:

- Ans
- ☒ 1. 0.5
 - ☒ 2. $\sqrt{2}$
 - ☒ 3. $1/\sqrt{2}$
 - ☒ 4. $2\sqrt{2}$

Question ID : 630680139573
Option 1 ID : 630680540345
Option 2 ID : 630680540342
Option 3 ID : 630680540343
Option 4 ID : 630680540344
Status : Answered
Chosen Option : 3

Q.52 The _____ has to be encoded with the magnitude to represent the integer completely.

- Ans
- ☒ 1. position
 - ☒ 2. sign information
 - ☒ 3. weight
 - ☒ 4. number of bits

Question ID : 630680103524
Option 1 ID : 630680402340
Option 2 ID : 630680402341
Option 3 ID : 630680402339
Option 4 ID : 630680402342
Status : Answered
Chosen Option : 2

Q.53 What is the typical response to a Non-Maskable Interrupt?

- Ans
- ☒ 1. The CPU switches to a different process.
 - ☒ 2. The system is halted.
 - ☒ 3. The CPU executes a specific interrupt service routine.
 - ☒ 4. The CPU continues its current task.

Question ID : 6306801046362
Option 1 ID : 6306804107563
Option 2 ID : 6306804107561
Option 3 ID : 6306804107564
Option 4 ID : 6306804107562
Status : Answered
Chosen Option : 3

Q.54 Maxterms is represented by _____ of all variables in complemented or direct form.

- Ans
- ☒ 1. product
 - ☒ 2. subtraction
 - ☒ 3. addition
 - ☒ 4. modular

Question ID : 630680103539
Option 1 ID : 630680402400
Option 2 ID : 630680402402
Option 3 ID : 630680402399
Option 4 ID : 630680402401
Status : Answered
Chosen Option : 1

Q.55 How does the bus grant signal propagated in the daisy chain method?

- Ans
- ☒ 1. In a circular fashion
 - ☒ 2. From the highest-priority DMA controller to the lowest-priority DMA controller
 - ☒ 3. From the CPU to the DMA controllers
 - ☒ 4. From the DMA controllers to the CPU

Question ID : 6306801046290
Option 1 ID : 6306804107276
Option 2 ID : 6306804107275
Option 3 ID : 6306804107274
Option 4 ID : 6306804107273
Status : Answered
Chosen Option : 2



Q.56 Two transformers operating in parallel will share the load in proportion to their kVA rating if which of the following conditions is satisfied?

Ans

✓ 1. Leakage impedances are inversely proportional to the respective kVA ratings.

✗ 2. Leakage impedances are directly proportional to the respective kVA ratings.

✗ 3. Leakage impedances are inversely proportional to the square of the respective kVA ratings.

✗ 4. Leakage impedances of the two transformers are equal.

Question ID : 63068061759

Option 1 ID : 630680238354

Option 2 ID : 630680238356

Option 3 ID : 630680238357

Option 4 ID : 630680238355

Status : Answered

Chosen Option : 1

Q.57 A single line-to-ground fault occurs in a three-phase system with an isolated neutral, where the line-to-neutral voltage is V kV. As a result of the fault, what will be the new voltage on the healthy phases?

Ans

✓ 1. $\sqrt{3}$ V kV

✗ 2. 3 V kV

✗ 3. $\frac{1}{\sqrt{3}}$ V kV

✗ 4. $\sqrt{2}$ V kV

Question ID : 6306801053205

Option 1 ID : 6306804135411

Option 2 ID : 6306804135412

Option 3 ID : 6306804135413

Option 4 ID : 6306804135410

Status : Answered

Chosen Option : 2

Q.58 What do you mean by tape cartridge?

Ans

✗ 1. A type of magnetic tape format

✓ 2. A small, portable container for magnetic tape

✗ 3. A device that reads and writes data on magnetic tape

✗ 4. A controller for magnetic tape

Question ID : 6306801046393

Option 1 ID : 6306804107686

Option 2 ID : 6306804107685

Option 3 ID : 6306804107687

Option 4 ID : 6306804107688

Status : Answered

Chosen Option : 3

Q.59 Given the hexadecimal number 7F2B4.D3, find its binary equivalent. Which of the following options correctly represents this binary number?

Ans

✗ 1. 0111 1111 0010 1011 0101. 1010 0010

✗ 2. 0111 1111 0010 1011 0010. 1010 1010

✗ 3. 0111 1111 0010 1011 0100 0000. 1101 0110

✓ 4. 0111 1111 0010 1011 0100. 1101 0011

Question ID : 6306801038453

Option 1 ID : 6306804076596

Option 2 ID : 6306804076598

Option 3 ID : 6306804076597

Option 4 ID : 6306804076595

Status : Answered

Chosen Option : 4

Consider the following Karnaugh Map for a Boolean function $f(A,B,C,D)$. Identify the statement that correctly describes a prime implicant.

$AB \backslash CD$	00	01	11	10
00	0	1	0	1
01	1	1	1	0
11	0	1	1	0
10	1	0	1	1

1. The group of minterms covering (00, 00), (00, 01), (01, 01), (01, 11), (11, 11), and (10, 11) is a prime implicant.
2. The group of minterms covering (00, 00), (01, 00), (11, 00), and (10, 00) is a prime implicant.
3. The group of minterms covering (01, 01), (01, 11), (11, 01), and (11, 11) is a prime implicant.
4. The group of minterms covering (01, 01), (01, 11), (11, 01), (11, 10) is a prime implicant.

Question
Option
Option
Option
Option
Status
Chosen Option

Q.61 How does the economiser improve the efficiency of the boiler?

1. By increasing the air temperature for combustion
2. By heating the feed water using heat from flue gases
3. By cooling the steam before it enters the turbine
4. By removing ash from the boiler

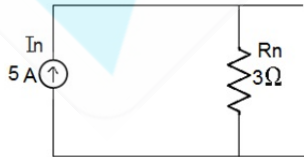
Question ID : 6306801053198
Option 1 ID : 6306804135384
Option 2 ID : 6306804135383
Option 3 ID : 6306804135382
Option 4 ID : 6306804135385
Status : Answered
Chosen Option : 2

Q.62 A 5000 KVA step down transformer, whose HV winding is rated for a voltage of 5000 V, has turns ratio of 2. Neglecting losses, estimate its HV side current when LV winding is connected with a load of 1 MW at 0.8 power factor lagging.

1. 500 A
2. 125 A
3. 250 A
4. 62.5 A

Question ID : 630680146450
Option 1 ID : 630680566684
Option 2 ID : 630680566682
Option 3 ID : 630680566683
Option 4 ID : 630680566685
Status : Answered
Chosen Option : 4

Q.63 Study the given circuit and find Thevenin voltage V_{th} and Thevenin resistance R_{th} .



1. 15 V, 3 Ω
2. 5 V, 3 Ω
3. 15 V, 1/3 Ω
4. 5 V, 1/3 Ω

Question ID : 630680117107
Option 1 ID : 630680453716
Option 2 ID : 630680453714
Option 3 ID : 630680453715
Option 4 ID : 630680453717
Status : Answered
Chosen Option : 1



Q.64 Consider a Double Sideband Suppressed Carrier Modulated waveform. If this waveform is input to ideal phase detector, then output of phase detector is:

- Ans
- ☒ 1. message signal waveform
 - ☒ 2. rectangular waveform
 - ☒ 3. constant
 - ☒ 4. triangular waveform

Question ID : 630680146387
Option 1 ID : 630680566432
Option 2 ID : 630680566431
Option 3 ID : 630680566430
Option 4 ID : 630680566433
Status : Answered
Chosen Option : 3

Q.65 The impulse response of a stable system has which of the following time responses?
a.Damped oscillations
b.Exponentially decreasing
c.Constant amplitude oscillations
d.Exponentially increasing oscillations

- Ans
- ☒ 1. (a), (b) and (c)
 - ☒ 2. (a) and (b)
 - ☒ 3. (b) and (c)
 - ☒ 4. (a), (b), (c) and (d)

Question ID : 63068093717
Option 1 ID : 630680363564
Option 2 ID : 630680363563
Option 3 ID : 630680363566
Option 4 ID : 630680363565
Status : Not Answered
Chosen Option : --

Q.66 Measurements using comparison methods in DC potentiometers are capable of a high degree of accuracy because:

- Ans
- ☒ 1. the result obtained depends upon the accuracy with which the voltage of the reference source is known and the actual deflection of the pointer
 - ☒ 2. a DC potentiometer makes use of a balance or null condition
 - ☒ 3. the result obtained depends upon the actual deflection of the pointer
 - ☒ 4. the result obtained depends upon the accuracy with which the voltage of the reference source is known

Question ID : 630680196607
Option 1 ID : 630680762729
Option 2 ID : 630680762730
Option 3 ID : 630680762727
Option 4 ID : 630680762728
Status : Answered
Chosen Option : 1

Q.67 What temperature change might occur before it is detected, if the dead zone in a certain pyrometer is 12.5% of span. The calibration is 400°C to 1000°C.

- Ans
- ☒ 1. 85°C
 - ☒ 2. 75°C
 - ☒ 3. 8°C
 - ☒ 4. 125°C

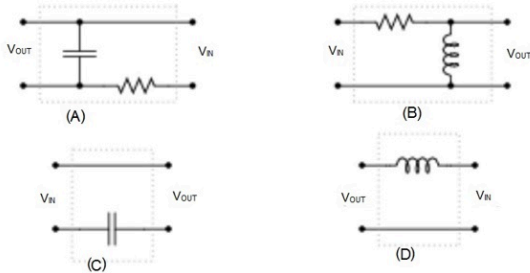
Question ID : 63068074516
Option 1 ID : 630680288222
Option 2 ID : 630680288224
Option 3 ID : 630680288225
Option 4 ID : 630680288223
Status : Answered
Chosen Option : 1

Q.68 Consider an RL series network. The output is taken across the inductor. The first and the last critical frequency of an RL driving point impedance function must be _____ and _____, respectively:

- Ans ☒ 1. a zero, a pole
☒ 2. a pole, a pole
☒ 3. a zero, a zero
☒ 4. a pole, a zero

Question ID : 630680146541
 Option 1 ID : 630680567049
 Option 2 ID : 630680567047
 Option 3 ID : 630680567046
 Option 4 ID : 630680567048
 Status : Answered
 Chosen Option : 4

Q.69 Study the given filter circuits, identify High-Pass filter circuits and select the correct option.



- Ans ☒ 1. B and C are High-Pass filter circuits.
☒ 2. A, B and D are High-Pass filter circuits.
☒ 3. B and D are High-Pass filter circuits.
☒ 4. A, B and C are High-Pass filter circuits.

Question ID : 630680117112
 Option 1 ID : 630680453735
 Option 2 ID : 630680453737
 Option 3 ID : 630680453736
 Option 4 ID : 630680453734
 Status : Answered
 Chosen Option : 1

Q.70 If a vector field $\vec{B}$ is solenoidal, which of the following is true?

- Ans ☒ 1. $\nabla \cdot \vec{B} = 0$
☒ 2. $\nabla \times \vec{B} = 0$
☒ 3. $\nabla \cdot \vec{B} \neq 0$
☒ 4. $\nabla \times \vec{B} \neq 0$

Question ID : 63068063875
 Option 1 ID : 630680246590
 Option 2 ID : 630680246591
 Option 3 ID : 630680246593
 Option 4 ID : 630680246592
 Status : Answered
 Chosen Option : 4

Q.71 Which of the following is NOT a correct statement?

- Ans ☒ 1. The performance of an SSB system using synchronous detection is equivalent to the performance of the both DSB and baseband systems.
☒ 2. Frequency modulation performance is always better than phase modulation scheme
☒ 3. Performance of DSB-LC is always inferior as compared to DSB-SC.
☒ 4. Figure of merit for conventional AM system is always less than or equal to one.

Question ID : 630680146385
 Option 1 ID : 630680566422
 Option 2 ID : 630680566424
 Option 3 ID : 630680566423
 Option 4 ID : 630680566425
 Status : Not Answered
 Chosen Option : --

Q.72 For a DC Chopper, supply voltage is 110 Volts and average load current is 30 Amp. For chopping frequency of 250 Hz and load resistance of $2\ \Omega$, Calculate ON time period of chopper.

- Ans
- ✓ 1. 2.18 msec
 - ✗ 2. 1.65 msec
 - ✗ 3. 3.36 msec
 - ✗ 4. 2.36 msec

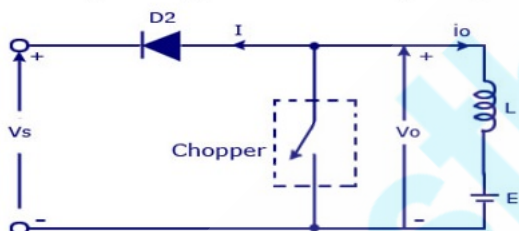
Question ID : 63068093445
 Option 1 ID : 630680362475
 Option 2 ID : 630680362477
 Option 3 ID : 630680362476
 Option 4 ID : 630680362478
 Status : Answered
 Chosen Option : 1

Q.73 Given that a point source of light emits 1.5 W of power, calculate the Poynting vector S at a distance of 0.30 m from the source.

- Ans
- ✗ 1. 13.28 W/m^2
 - ✗ 2. 1.77 W/m^2
 - ✗ 3. 4.42 W/m^2
 - ✓ 4. 1.33 W/m^2

Question ID : 6306801050154
 Option 1 ID : 6306804122964
 Option 2 ID : 6306804122962
 Option 3 ID : 6306804122963
 Option 4 ID : 6306804122961
 Status : Answered
 Chosen Option : 1

Q.74 Which type of chopper is shown in the given figure?



- Ans
- ✗ 1. Type A chopper
 - ✗ 2. Type E chopper
 - ✗ 3. Type D chopper
 - ✓ 4. Type B chopper

Question ID : 630680119911
 Option 1 ID : 630680464710
 Option 2 ID : 630680464711
 Option 3 ID : 630680464713
 Option 4 ID : 630680464712
 Status : Answered
 Chosen Option : 1

Q.75 In a non-ideal transformer, the electrical energy from one set of winding is transferred to another set of winding without affecting any change in which of the following parameters?

- Ans
- ✗ 1. Current
 - ✓ 2. Frequency
 - ✗ 3. Power
 - ✗ 4. Voltage

Question ID : 63068061738
 Option 1 ID : 630680238272
 Option 2 ID : 630680238270
 Option 3 ID : 630680238273
 Option 4 ID : 630680238271
 Status : Answered
 Chosen Option : 2

Q.76 Match the following oscillators with the expected waveforms.

Oscillator	Expected waveform
1. LC oscillator	A. Sinusoidal
2. RC oscillator	B. Triangular
3. Crystal oscillator	C. Square wave

- Ans
- ☒ 1. 1 - B, 2 - C, 3 - A
 - ☒ 2. 1 - C, 2 - A, 3 - B
 - ☒ 3. 1 - B, 2 - A, 3 - C
 - ☒ 4. 1 - A, 2 - B, 3 - C

Question ID : 630680117051
Option 1 ID : 630680453493
Option 2 ID : 630680453491
Option 3 ID : 630680453490
Option 4 ID : 630680453492
Status : Answered
Chosen Option : 2

Q.77 Any binary signed number is recognised by its _____.

- Ans
- ☒ 1. Byte
 - ☒ 2. LSB
 - ☒ 3. MSB
 - ☒ 4. Nibble

Question ID : 630680176331
Option 1 ID : 630680683554
Option 2 ID : 630680683553
Option 3 ID : 630680683552
Option 4 ID : 630680683555
Status : Answered
Chosen Option : 3

Q.78 Compute the DFT spectrum of $\{x(0) = 3, x(1) = 4, x(2) = 3, x(3) = -2\}$.

- Ans
- ☒ 1. $\{X(0) = 8, X(1) = -j6, X(2) = 4, X(3) = j6\}$
 - ☒ 2. $\{X(0) = 8, X(1) = 4 - j6, X(2) = 0, X(3) = j6\}$
 - ☒ 3. $\{X(0) = 8, X(1) = 6, X(2) = 4, X(3) = 6\}$
 - ☒ 4. $\{X(0) = 3, X(1) = -j4, X(2) = 3j, X(3) = 2\}$

Question ID : 630680742616
Option 1 ID : 6306802910774
Option 2 ID : 6306802910775
Option 3 ID : 6306802910776
Option 4 ID : 6306802910777
Status : Not Answered
Chosen Option : --

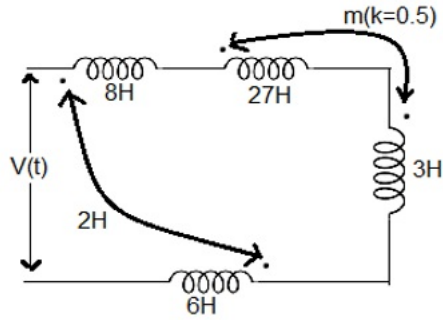
Q.79 Which of the following is a characteristic of magnetically hard materials?

- Ans
- ☒ 1. High BH product
 - ☒ 2. Low retentivity
 - ☒ 3. Low magnetic reluctance
 - ☒ 4. Low coercivity

Question ID : 6306801046120
Option 1 ID : 6306804106566
Option 2 ID : 6306804106565
Option 3 ID : 6306804106567
Option 4 ID : 6306804106564
Status : Answered
Chosen Option : 1



Q.80 Study the given circuit and find L_{eq} (equivalent inductance).



- Ans
- ☒ 1. 44
 - ☒ 2. 46
 - ☒ 3. 57
 - ☒ 4. 50.5

Question ID : 630680117117
 Option 1 ID : 630680453754
 Option 2 ID : 630680453757
 Option 3 ID : 630680453755
 Option 4 ID : 630680453756
 Status : Answered
 Chosen Option : 2

Q.81 Series connection method of Q-meter is employed for measurement of:

- Ans
- ☒ 1. low-impedance components
 - ☒ 2. high-impedance components
 - ☒ 3. high value resistors
 - ☒ 4. small capacitors

Question ID : 63068067117
 Option 1 ID : 630680259301
 Option 2 ID : 630680259299
 Option 3 ID : 630680259298
 Option 4 ID : 630680259300
 Status : Answered
 Chosen Option : 1

Q.82 The data on the tracks of magnetic disks is written as:

- Ans
- ☒ 1. tiny magnetic spots
 - ☒ 2. down or up
 - ☒ 3. 0 or 1
 - ☒ 4. low or high voltage



Question ID : 630680192609
 Option 1 ID : 630680747000
 Option 2 ID : 630680746998
 Option 3 ID : 630680746997
 Option 4 ID : 630680746999
 Status : Answered
 Chosen Option : 1

Q.83 What is the formula for the capacitance per unit length (C) of a coaxial cable, where a is the radius of the inner conductor and b is the radius of the outer conductor?

- Ans
- ☒ 1. $\frac{\epsilon_0 \epsilon_r}{\pi} \ln\left(\frac{b}{a}\right)$
 - ☒ 2. $\frac{2\pi\epsilon_d}{\ln\left(\frac{b}{a}\right)}$
 - ☒ 3. $\frac{\epsilon_0 \epsilon_r \pi}{\ln\left(\frac{b}{a}\right)}$
 - ☒ 4. $\frac{\epsilon_d}{\pi \ln\left(\frac{b}{a}\right)}$

Question ID : 6306801050161
 Option 1 ID : 6306804122989
 Option 2 ID : 6306804122990
 Option 3 ID : 6306804122991
 Option 4 ID : 6306804122992
 Status : Answered
 Chosen Option : 3

Q.84 The Q factor of an RC series circuit is:

Ans

✗ 1. $2\pi f \frac{R}{C}$

✓ 2. $\frac{1}{2\pi fCR}$

✗ 3. $2\pi fCR$

✗ 4. $\frac{2\pi fC}{R}$

Question ID : 630680146535
Option 1 ID : 630680567025
Option 2 ID : 630680567024
Option 3 ID : 630680567023
Option 4 ID : 630680567022
Status : Answered
Chosen Option : 3

Q.85 What is the difference between SRAM and DRAM?

Ans

✗ 1. SRAM is used for main memory, while DRAM is used for cache

✓ 2. SRAM is faster but more expensive than DRAM.

✗ 3. SRAM is slower but cheaper than DRAM.

✗ 4. SRAM is non-volatile, while DRAM is volatile.

Question ID : 6306801046385
Option 1 ID : 6306804107656
Option 2 ID : 6306804107653
Option 3 ID : 6306804107654
Option 4 ID : 6306804107655
Status : Answered
Chosen Option : 3

Q.86 The positive and negative numbers are separated in the number system by using

Ans

✗ 1. 0

✗ 2. LSB

✗ 3. 1

✓ 4. MSB

Question ID : 630680103527
Option 1 ID : 630680402353
Option 2 ID : 630680402351
Option 3 ID : 630680402354
Option 4 ID : 630680402352
Status : Answered
Chosen Option : 4

Q.87 Which of the following is the solution of the differential equation $x^2 \frac{d^2y}{dx^2} + 5x \frac{dy}{dx} + 4y = x \log x$?

Ans

✗ 1. $y = \frac{C_1 + C_2 \log x}{x^2} - \frac{x}{9} \left(\log x + \frac{2}{3} \right)$

✗ 2. $y = \frac{C_1 + C_2 \log x}{x^2} + \frac{x}{9} \left(\log x + \frac{2}{3} \right)$

✗ 3. $y = \frac{C_1 + C_2 \log x}{x^2} - \frac{x}{9} \left(\log x - \frac{2}{3} \right)$

✓ 4. $y = \frac{C_1 + C_2 \log x}{x^2} + \frac{x}{9} \left(\log x - \frac{2}{3} \right)$

Question ID : 630680208716
Option 1 ID : 630680809097
Option 2 ID : 630680809098
Option 3 ID : 630680809096
Option 4 ID : 630680809095
Status : Not Answered
Chosen Option : --



Q.88 If the modulation index of an AM is changed from 0 to 1 for single tone modulation, the transmitted power:

Ans

- 1. reduces by 50%
- 2. increases by 100%
- 3. increases by 50%
- 4. remains unchanged

Question ID : 630680120816

Option 1 ID : 630680468238

Option 2 ID : 630680468240

Option 3 ID : 630680468239

Option 4 ID : 630680468241

Status : Answered

Chosen Option : 1

Q.89 Consider a square matrix A which is NOT symmetric. Which of the following operations will lead to a symmetric matrix?

Ans

- 1. $A^T A A^T$
- 2. $A^2 A^T$
- 3. $A - A^T$
- 4. $A + A^T$

Question ID : 630680103926

Option 1 ID : 630680403922

Option 2 ID : 630680403921

Option 3 ID : 630680403920

Option 4 ID : 630680403919

Status : Not Answered

Chosen Option : --

Q.90 For a time – varying magnetic field, the equation $\nabla \cdot \vec{B} = 0$ implies that:

Ans

- 1. the magnetic field density changes with time
- 2. the magnetic flux through a closed surface is equal to the current passing through it
- 3. the magnetic field is a source of electric current
- 4. the magnetic field lines have no sources or sinks

Question ID : 6306801038042

Option 1 ID : 6306804074972

Option 2 ID : 6306804074973

Option 3 ID : 6306804074971

Option 4 ID : 6306804074970

Status : Answered

Chosen Option : 2

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